



RAKTSETU E-BLOOD BANK

A Smart Healthcare Technology

[Software Requirements Specifications]

Version 1.0



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Course : Diploma [CSE-AIML]

Lab Type : PBL (Problem Based Learning)

Date : 24th February , 2026

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to provide a comprehensive description of the **RaktSetu** platform. This document defines the functional and non-functional requirements to ensure the seamless bridge between blood donors and those in urgent medical need. It serves as the primary benchmark for the development team, UI/UX designers, and medical stakeholders.

KEY OBJECTIVES OF THIS SPECIFICATIONS :

- **Establish a Single Source of Truth:** To ensure that all four stakeholders (Donors, Recipients, Hospitals, and NGOs) operate within a synchronized logic, eliminating communication gaps during the "Golden Hour."
- **Define Legal & Regulatory Compliance:** To outline the strict protocols for handling sensitive Government Identity documents (Aadhar, PAN, etc.) and ensuring data privacy aligns with national health-tech standards.
- **Standardize Technical Architecture:** To provide clear specifications for the **Bento Grid** UI and the high-performance back-end, ensuring the system remains stable under the high-pressure demands of emergency broadcasts.
- **Facilitate Scalability:** To serve as the foundation for future expansions, such as AI-driven shortage prediction and native mobile application development, ensuring the core logic is robust enough to grow.
- **Ensure Operational Accountability:** To define the verification "State Machine" for every blood request and donor match, creating a clear audit trail for medical and legal transparency.
- **Minimize Technical Debt:** By providing detailed functional requirements early, this document aims to reduce the need for costly redesigns during the implementation phase.

1.2 Document Conventions

This document follows the **IEEE 830-1998** standard for Software Requirements Specifications. The following conventions are used throughout to ensure clarity and consistency:

1.2.1 Typography and Formatting

- **Bold Face:** Used to highlight Key **Stakeholders**, **Primary Features**, and **User Interface Actions** (e.g., "The **Hospital** clicks the **Verify** button").
- *Italics:* Reserved for *Technical Terminology*, *External API names*, or *Specific UI Labels* (e.g., the *Bento Grid* layout or *Google Maps API*).
- **Monospace Text:** Used for specific code variables, database field names, or file paths (e.g., `user_id` or `document_status`).

1.2.2 Requirement Priority Grading

Each requirement is assigned a **priority level** to guide the development phases:

- **High (Critical):** Requirements essential for the "Golden Hour" emergency response and core life-saving functionality. Must be implemented in the MVP (Minimum Viable Product).
- **Medium (Important):** Features that enhance user engagement, NGO operations, and reporting.
- **Low (Optional):** Features intended for future roadmaps, such as AI-driven analytics or third-party integrations.

1.2.3 Requirement Tagging

To simplify tracking, every requirement will follow a unique ID format:

- **FR-XX:** Functional Requirement (e.g., **FR-01: Identity Verification**).
- **NFR-XX:** Non-Functional Requirement (e.g., **NFR-01: System Latency**).

1.2.4 Visual Standards*

- **Color Coding:** References to the color palette strictly follow the **RaktSetu** brand guidelines (Deep Teal: **#0d9488**).
- **Diagram Standards :** All flowcharts and system models utilize standard UML (Unified Modeling Language) notation.

1.3 Intended Audience

This document is intended for :

- a) Software Developers & Architects:** To understand the functional logic, database schema, and the "Bento Grid" UI implementation requirements.
- b) Project Managers & Scrum Masters:** To break down the features into development sprints and track project milestones.
- c) Medical & Hospital Administrators:** To evaluate how the system handles blood inventory and verifies patient emergency requests.
- d) Government & Regulatory Authorities:** To audit the security of sensitive document handling (Aadhar, PAN, etc.) and ensure compliance with health data privacy laws.
- e) NGO Coordinators & Volunteers:** To understand the tools available for organizing blood drives and tracking community impact.
- f) Quality Assurance (QA) & Testing Team:** To create test cases based on the functional requirements, ensuring the "Golden Hour" broadcast works without failure.
- g) Legal & Compliance Officers:** To review the "Data Lifecycle" and ensure that the digital storage of identity documents meets national legal standards.
- h) UI/UX Designers:** To implement the Bento Grid layout and ensure the color palette and typography match the established medical-grade brand identity.

1.4 Project Scope

RaktSetu is an intelligent , real – time blood donation and management ecosystem. It aims to eliminate the “Life - Gap” between blood availability and emergency requirements.

- The system acts as a dynamic coordinator between four critical stakeholders : **Donors , Recipients , Hospitals & NGOs.**
- **Verified Identity Gateway:** Integration of a multi-document verification module that supports **Aadhar, PAN, and License** cards, ensuring that every user (Donor/Recipient) is a legally traceable entity.
- **Geofencing & Proximity Logic:** Implementation of a location-based service that limits "Urgent Requests" to donors within a specific radius (e.g., **10–20km**) to ensure blood arrives within the **Golden Hour.**
- **Automated Impact Certification:** System-generated, NGO-verified donation certificates and "Impact Reports" that can be shared on professional networks (like LinkedIn) to encourage a culture of donation.
- **Emergency Escalation Protocol:** A tiered notification system where if a "Critical" request isn't accepted within **15 minutes**, the broadcast radius automatically expands and notifies registered NGOs for manual intervention.
- **Privacy-Preserving Communication:** An in-app messaging or masked-calling feature that allows Hospitals to coordinate with Donors without revealing the donor's private phone number to the general public.
- **Fraud Detection & Blacklist Management:** A monitoring layer to identify and block "fake requests" or accounts that attempt to commercialize blood donation, maintaining the platform's ethical integrity.

1.5 Definitions , Acronyms , and Abbreviations

This section provides a glossary of terms used throughout the SRS to ensure a common understanding among developers, medical staff, and government authorities.

Term	Definition
ABO/Rh	The standard classification system for blood types (A, B, AB, O) and Rhesus factor (+/-).
Aadhar/KYC	Government-issued ID and the "Know Your Customer" process used to verify the legal identity of donors and recipients.
Bento Grid	A UI design pattern that organizes information into modular, rectangular "tiles" for high-pressure data consumption.
Golden Hour	The critical 60-minute window following a traumatic injury where rapid blood transfusion is most likely to save a life.
LIMS	Laboratory Information Management System; the internal software used by hospitals to manage blood bank inventory.
RBAC	Role-Based Access Control; a security protocol that restricts system access to authorized users based on their role (e.g., Hospital Admin vs. NGO Volunteer).
AES-256	Advanced Encryption Standard with a 256-bit key; the high-level security protocol used by RaktSetu to protect uploaded legal documents.
Impact Counter	A gamified metric that tracks the total number of lives saved or influenced by a specific donor's contributions.
Cooling-off Period	The mandatory 90-day waiting interval between whole blood donations to ensure donor recovery and health safety.
Geolocation (GPS)	The use of global positioning systems to find and alert the nearest compatible donors to an emergency site.

2. Overall Description

2.1 Product Perspective

RaktSetu is a standalone, web-based ecosystem designed to bridge the gap between fragmented hospital blood banks and the general public.

- **Integration Layer:** The system is designed to interface with Hospital **LIMS** (Laboratory Information Management Systems) to read live inventory levels without replacing the hospital's internal medical record-keeping.
- **Government Alignment:** It serves as a verification gateway that connects with official identity records to ensure that every **Aadhar** or **PAN** card uploaded is authentic, maintaining a high-trust environment.
- **Modular Interface:** Utilizing the **Bento Grid** UI, the platform provides different "views" based on the user class, ensuring that a Donor sees a simple call-to-action while a Hospital Admin sees a complex data dashboard.

2.2 Product Functions

The high-level functions of **RaktSetu** are divided into the following primary workflows:

- **Verified Onboarding (KYC Orchestration):** The system manages the collection, encryption, and approval of government-issued IDs for all stakeholders to prevent fraudulent requests.
- **Intelligent Matchmaking:** An automated engine that cross-references the **ABO/Rh** blood type of a recipient with the real-time **GPS** location of eligible donors within a specified radius.
- **Emergency Broadcasting:** A "One-Click" emergency trigger for hospitals that sends high-priority notifications across multiple channels (SMS, Push, Web) during the **Golden Hour**.

- **Donor Lifecycle Management:** Tracking the donor from the moment they accept a request to the physical donation, followed by enforcing the **90-day cooling-off period**.
- **Resource Mapping:** A real-time map for NGOs and Recipients to see which nearby hospitals have specific blood units in stock, preventing wasted time during emergencies.

2.3 User Classes and Characteristics

2.3.1 Donors (The “ Life – Savers ”) :

- **Persona :** Individuals willing to donate blood voluntarily.
- **Characteristics :** Usually mobile – first users who value speed and recognition.
- **System Interaction (Gamified Experience) :**
 - Access to a **Personal Impact Dashboard** showing lives saved.
 - Real – Time notifications for nearby “ Golden Labour ” emergencies based on their current **GPS** location.
 - Automated eligibility tracking that counts down the cooling – off period with the help of recent blood type donated as given below :
 - **Whole Blood :** cooldown period of **56 days / 6 times per year**.
 - **Platelets :** cooldown period of **7days / 24times – year**.
 - **Plasma :** cooldown period of **28days / 13times – year**.
 - **Double Red Cells / Power Red :** cooldown period of **112days / 3-4 times – year**.
 - In the Situation of **Emergencies** , the donor can donate again by using emergency bypass button given in dashboard to the user(**Donor**).
 - Earning digital badges and NGO-verified certificates for social sharing.

2.3.2 Recipients (The “ Emergency Command Centre ”)

- **Persona :** Family members or patients in high-stress , time-critical situations.
- **Characteristics :** Users who need a “Zero – Friction” interface with absolute minimal form – filling.

- **System Interaction :**
 - Simplified “One-Tap-Request” interface to broadcast blood needs.
 - Live tracking of “Incoming Donors” (Similar to a ride – sharing app) once a donor accepts the request .
 - Direct , secure communication channel with verified hospitals.

2.3.3 Hospitals (Inventory & Verification) :

- **Persona :** Pathologists, blood bank – technicians , and hospital administrators .
- **Characteristics :** Professional users who require high data density and legal accountability .
- **System Interaction :**
 - **Verification Portal :** Reviewing and approved uploaded government IDs (**Aadhar , PAN**).
 - **Inventory Management :** Updating live stock levels to prevent “Resources Wastage” .
 - **Check – in Logic :** Confirming the physical arrival of a donor and updating the system status to “Donation Successful”.

2.3.4 NGOS (Community Outreach) :

- **Persona :** Social organization and community leaders.
- **Characteristics :** Power users who manage bulk data and large – scale events.
- **System Interaction :**
 - **Event Coordinator :** Planning and broadcasting scheduled blood donation camps.
 - **Reporting Engine :** Exploring data on regional blood shortages and donor turnout for government reporting.
 - **Volunteer Management :** Assigning local volunteers to assist donors at hospital sites.

2.4 Operating Environment

RaktSetu is designed as a cloud – native , responsive web ecosystem. To ensure 100% availability during medical emergencies , it must operates under the following conditions :

- **Platform Compatibility** : A cross – platform web application accessible via modern browsers including Google Chrome (v100+) , Safari , Firefox